

Assignment 4 Report

Reviewer: Megan Chau (mhc23) Reviewee: Samantha Gan (ssg46)

March 31, 2026

In this report, I reviewed Samantha Gan's implemented code. I looked through three files that she implemented: GameScreen.java, EndScreen.java, and House.java. Authorship was confirmed with git blame. The four code blames are identified below.

1 Smell 1: Unnecessary or useless variables

Files: GameScreen.java, lines 1003 to 1004 **Commit:** b3df3c0

```
bikeParked = true;
isOnBike = false;
```

bikeParked is always isOnBike. These two booleans track the same fact from two different angles, and they can diverge. For example, activateMap() sets bikeParked = false; isOnBike = true; correctly, but any future path that forgets to synchronize will cause an issue. Replace both with a single isOnBike; bikeParked becomes if isOnBike.

2 Smell 2: Data Clumps

File: EndScreen.java, lines 30 to 43 **Commit:** 50d9c58

```
private static final float DIGIT_W    = 150f;
private static final float DIGIT_H    = 140f;
private static final float DIGIT_X    = 348f;
private static final float DIGIT_Y    = 460f;
private static final float DIGIT_GAP  = 139f;

private static final float TIME_W      = 50f;
private static final float TIME_H      = 50f;
private static final float TIME_X      = 450f;
private static final float TIME_Y      = 328f;
private static final float TIME_GAP    = 35f;
```

The file includes a lot of float literals assigned to constant names. Taken together, the constants DIGIT_X, DIGIT_Y, DIGIT_W, DIGIT_H, DIGIT_GAP and their TIME_ counterparts form a data clump. They are used together, and describe the same concept but are stored as separate unrelated fields.

3 Smell 3: Code Duplication

File: EndScreen.java, lines 106 to 124 (unrelated comments and code removed for simplicity) **Commit:** 50d9c58

```
int[] scoreDigits = {
    (s / 1000) % 10,
    (s / 100)  % 10,
    (s / 10)   % 10,
    s         % 10
};
for (int i = 0; i < 4; i++) {
    batch.draw(digits[scoreDigits[i]], DIGIT_X + i * DIGIT_GAP, DIGIT_Y, DIGIT_W, DIGIT_H);
}
int ss = time % 60;
int[] timeDigits = { mm / 10, mm % 10, ss / 10, ss % 10 };
for (int i = 0; i < 4; i++) {
    float offset = (i < 2) ? i * TIME_GAP : i * TIME_GAP + TIME_COLON_W;
    batch.draw(timeNumbers[timeDigits[i]], TIME_X + offset, TIME_Y, TIME_W, TIME_H);
}
```

The score-digit array and the time-digit array are both computed inside render() using identical arithmetic operations (/ 10000, % 10, etc.). This pattern should be a shared helper.

4 Smell 4: Dead Code

Files: House.java, line 3 Commit: 50d9c58

```
import com.badlogic.gdx.Gdx;
```

Methods from this import are never used. Unnecessary imports can cause confusion and should be cleaned up.

Refactoring (Based on Lucas's Review)

After receiving Lucas's code review, I refactored each identified code smell as follows:

Smell 1: Code Duplication Across Two Screen Classes Commit: 77e81fd

I introduced a new helper method, ScreenAudioHelper.java, to simplify code and to extract four identical lines from both InstructionsScreen and SettingsScreen methods into loadAndPlayEntrySound() and load(). The volume literal 0.8f is now a single named constant UI_SOUND.VOLUME. This adds more clarity and simplicity to the classes.

Smell 2: Data Clumps for the Toggle Button Hit Zone Commit: 5062325

I added a new file, HitZone that wraps the four toggle-button floats into a value object with a contains(Vector2) method. The four-part boolean in SettingsScreen.render() becomes a single readable TOGGLE_ZONE.contains(touch). This groups like variables together into one object, making the code more tidy and less complex.

Smell 3: Constants Updated Without Updating the Accompanying Comment Commit: 29afa63

In EndScreen.java, I updated the comments for SCORE_LAYOUT and TIME_LAYOUT to describe these variables more accurately. I learned that when changing code, it is also important to change the comments related to the change in order to prevent any confusion for my teammates or readers.

```
/**
 * Score box: 4 large digits at the middle top part of the endScreen, to represent the score
 * x = x-cor, y = y-cor, w = width, h = height, gap = spacing between digits
 */
...
/**
 * Time row: 4 smaller digits with different appearance for representation of time
 * x = x-cor, y = y-cor, w = width, h = height, gap = spacing between digits
 */
```

Smell 4: Sound Objects Not Disposed Commit: 77e81fd

I added toggleSound.dispose() and escSound.dispose() inside instructionsScreen's and SettingsScreen's dispose() method. I later removed escSound.dispose() in both files later (since I refactored this smell first and the 1st smell last), and realized that when escSound.dispose() is called, it does not play escSound when ESC is pressed.

```
public void dispose() {
    toggleSound.dispose();
    escSound.dispose(); //omitted in actual code since it causes error in sound
}
```